

Machine Learning for Quantified Self

Deadline 2

Misha Markovits - 2714369, Noëlla Hilgen - 2889966, and Luka Zaitch - 2806898

Vrije Universiteit, Amsterdam, The Netherlands

1 Feature Engineering

The goal of this step is to transform the raw multi-sensor time series into window-based feature vectors that encode both short-term and longer-term dynamics. We mostly followed the approach from Chapter 4 of the book [4], and added some additional features that we will describe later. Our dataset consists of accelerometer and gyroscope measurements from a smartphone in the right jeans pocket, accelerometer and gyroscope from a smartwatch on the left wrist, and heart rate (HR) measurements, recorded while the participant listens to no music, soft techno, or hard techno. Our question is whether these features allow us to find a difference between the three music conditions based on movement and physiological responses.

1.1 Windowing strategy

After preprocessing and resampling, we segment the merged sensor stream into sliding windows. Each window is defined by a duration W (in seconds) and an overlap O (in seconds) between consecutive windows. Rather than using a single global configuration, our framework allows each feature group to be assigned its own independent windowing parameters (W_f, O_f) . This is motivated by the different temporal scales at which signals operate: fast-varying channels such as raw accelerometry benefit from shorter, higher-resolution windows, while slower signals such as heart rate require longer windows to capture meaningful variation. All per-group windows are synchronised to a common reference time grid so that features computed at different scales can be fused at the segment level without misalignment.

The specific values of (W_f, O_f) will be determined empirically once the full dataset has been collected. Candidate configurations are evaluated on a held-out validation set guided by two criteria: prediction accuracy and computational cost. This data-driven approach ensures the windowing strategy reflects the actual properties of the collected data rather than assumptions made a priori.

1.2 Sensor channels and magnitude features

To improve robustness to device orientation and small shifts in placement during movement, we extend the raw sensor channels with vector magnitudes [10]. For

each accelerometer and gyroscope, we derive an additional magnitude channel defined as

$$\text{mag}(t) = \sqrt{x(t)^2 + y(t)^2 + z(t)^2} \quad (1)$$

and treat it as a separate column in the feature pipeline. These magnitude signals directly capture overall movement intensity and angular velocity in each body location and are less sensitive to rotations of the phone or the watch than the individual axes, which is relevant in our setting where we may walk with slightly different device orientations.

1.3 Time-domain features

For every base window and each sensor channel, we compute a set of time-domain features that summarize the local behavior of the signal, as shown in Section 4.1 of the book [4]. The implemented feature set includes:

- Mean, standard deviation and variance, capturing the local level and variability of the signal,
- Minimum, maximum, median and peak-to-peak range, describing the distribution and dynamic range of values within the window,
- Root-mean-square (RMS) and the absolute integral of the signal, which are intensity or energy measures over the window. RMS is normalized by window length and the absolute integral reflects the total absolute area under the curve. These two features may be correlated, so one of them may later be removed,
- A linear slope, obtained by fitting a straight line to the samples via least-squares, summarizing local trends,
- Zero-crossing rate and mean-crossing rate, counting how often the signal crosses zero or its own mean, characterizing oscillatory behavior and short-term fluctuations in the signal [7]. Note that the zero-crossing feature is not very important for HR data, as these are strictly positive.

These time-domain statistics are intended to be informative in our setting because they quantify the strength and variability with which we move under different music conditions, and how quickly these movements change within a window. For the pocket and wrist sensors, intensity and variability measures may separate periods with little movement from periods with more vigorous motion, while crossing rates relate to the rhythm and jitter of the movements. For HR, window-level averages and trends describe slower physiological responses, adding extra information on top of the faster motion signals.

1.4 Frequency-domain features

To capture periodic structure introduced by music (e.g. beat-synchronous movements), we also derive features from the frequency domain using FFT-based power spectral density (PSD) estimates, following the Fourier-based approach

in Section 4.2 [4] and the lecture. For each window and sensor channel we estimate the one-sided PSD using Welch’s method [9] when enough samples are available, and fall back to FFT for very short segments. From the resulting spectrum we compute:

- The dominant frequency, which is the frequency bin with the highest spectral power, reflecting the main periodicity within the window,
- A spectral centroid, which is the frequency-weighted average over the spectrum, summarizing the balance between low and high frequencies,
- Power spectral entropy, obtained by normalizing squared amplitudes to a probability distribution over frequencies and applying the standard entropy formula, which measures how concentrated the spectrum is in a few peaks versus spread across many frequencies.

In addition, we compute two other spectral descriptors. These are common in audio analysis, but are defined on the power spectrum and can therefore also be applied to sensor data [4, 6, 1]:

- Spectral rolloff, defined as the frequency below which a fixed fraction of the total spectral energy is contained, providing another indicator of how quickly energy decays towards higher frequencies [6, 2],
- Band energy ratios, where we integrate the PSD over several predefined frequency bands and divide by the total energy, as an extension of the book [4, 1, 5].

These frequency-domain features are intended to capture periodic movement patterns and the distribution of spectral energy in the accelerometer and gyroscope magnitudes, which may differ across the music conditions. In particular, changes in dominant frequency, spectral balance, and spectral entropy may reflect differences in how regular, intense, or rhythmically structured the walking motion is while listening to no music, soft techno, or hard techno.

1.5 Statistical and complexity features

We also wanted to go beyond the features the book explained, so we decided to compute additional statistical features that describe the shape and complexity of the signal within each window. Specifically, for each sensor channel we include:

- Skewness and kurtosis, capturing asymmetry and tail heaviness that are not visible in the mean and variance,
- A histogram-based Shannon entropy of the value distribution, reflecting how concentrated or spread the amplitudes are over their range [8],
- Hjorth mobility and Hjorth complexity, two classical time-series descriptors that quantify the overall frequency information and irregularity of the signal. Mobility is defined via the variance of the first derivative relative to the original variance, while complexity compares the mobility of the derivative to the original mobility [3],

- Interquartile range (IQR), which is the difference between the 75th and 25th percentiles, capturing the spread of the data.

These features are relevant in our setting because they can distinguish more regular, smooth movements (e.g. gentle sway with soft techno) from more high-energy movement patterns (e.g. fast movements to hard techno) even when simple averages or variances are similar.

1.6 Cross-sensor features

To exploit the fact that we measure motion at two body locations (pocket and wrist) plus HR, we also derive cross-sensor features. For each window we compute, for example, the difference and ratio between the mean and standard deviation of pocket and wrist accelerometer magnitudes, as well as the correlation between these magnitudes. These features encode how synchronized or decoupled the two body parts move under different music conditions and may help capture differences in movement style.

2 Classical machine learning models and train/test setup

Model selection. We evaluate three classical classifiers that span different inductive biases:

1. **Random Forest** an ensemble of decision trees trained on bootstrapped samples. It handles non-linear decision boundaries and feature interactions naturally, provides built-in feature importance, and is robust to irrelevant features and outliers.
2. **SVM** maps the feature space into a higher-dimensional Reproducing Kernel Hilbert Space, capturing non-linear separations that a linear model would miss.
3. **XGBoost** a gradient-boosted tree ensemble that sequentially corrects residual errors. It often achieves the highest raw accuracy on tabular benchmarks, at the cost of additional hyperparameters that require careful tuning.

Train/test setup with block-based splitting. Sliding window feature extraction creates strong temporal dependencies between adjacent windows; a random split would leak information from training windows into the test set, inflating performance estimates. To prevent this, we adopt a **block-based split**:

1. The raw time series is divided into N non-overlapping blocks (we use $N = 10$).
2. Blocks are shuffled and independently windowed, so that no window in the training set shares raw samples with any window in the test set.
3. The first 80 % of blocks form the validation pool (used for hyperparameter tuning and cross-validation); the remaining 20 % are held out for final evaluation. (This will probably still change to 100% used for validation, and a separate never-seen dataset serves as the test data.)

Table 1. Fixed hyperparameters used for each classical model.

Model	Hyperparameter	Value
Random Forest	<code>n_estimators</code>	100
	<code>max_depth</code>	10
SVM (RBF)	<code>kernel</code>	<code>rbf</code>
	<code>C</code>	1.0
	<code>gamma</code>	<code>scale</code>
XGBoost	<code>n_estimators</code>	100
	<code>learning_rate</code>	0.1
	<code>max_depth</code>	6

Hyperparameters and evaluation. Each model is trained with a fixed set of sensible defaults (listed in Table 1) rather than tuned via an exhaustive grid search. Within the training pool we perform k -fold cross-validation with folds defined by block assignment, ensuring all windows from the same original block remain in the same fold. This gives an unbiased estimate of generalisation performance on the training data. The model is then retrained on the full training pool and evaluated once on the held-out test set. The results across all three models are compared to identify the most effective approach for this task, with the option to perform more extensive hyperparameter tuning in a follow-up iteration. Because data collection is currently ongoing, we omitted the results from this report.

References

1. Bogdanov, D., Wack, N., Gómez Gutiérrez, E., Gulati, S., Herrera, P., Mayor, O., Roma, G., Salamon, J., Zapata, J.R., Serra, X.: *Essentia: An audio analysis library for music information retrieval*. In: *Proceedings of the 14th International Society for Music Information Retrieval Conference (ISMIR)*. pp. 493–498 (2013)
2. Eyben, F.: *Real-time Speech and Music Classification by Large Audio Feature Space Extraction*. Springer (2016). <https://doi.org/10.1007/978-3-319-27299-3>
3. Hjorth, B.: Eeg analysis based on time domain properties. *Electroencephalography and Clinical Neurophysiology* **29**(3), 306–310 (1970). [https://doi.org/10.1016/0013-4694\(70\)90143-4](https://doi.org/10.1016/0013-4694(70)90143-4)
4. Hoogendoorn, M., Funk, B.: *Machine Learning for the Quantified Self: On the Art of Learning from Sensory Data*, Cognitive Systems Monographs, vol. 35. Springer, Cham (2018). <https://doi.org/10.1007/978-3-319-66308-1>
5. Nicolai, M., et al.: Discriminative ratio of spectral power and relative power features derived via frequency-domain model ratio with application to seizure prediction. *IEEE Transactions on Biomedical Engineering* **66**(8), 2263–2274 (2019). <https://doi.org/10.1109/TBME.2019.2911630>
6. Peeters, G.: *A large set of audio features for sound description (similarity and classification) in the cuidado project*. Tech. Rep. IRCAM-02, IRCAM (2004), technical report

7. Prabhu, N., Asnodkar, A., Kenkre, R.: Multiclass support vector machine for music genre classification. *International Journal of Computer Applications* **107**(19), 15–17 (2014)
8. Shannon, C.E.: A mathematical theory of communication. *Bell System Technical Journal* **27**(3), 379–423 (1948)
9. Welch, P.D.: The use of fast fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms. *IEEE Transactions on Audio and Electroacoustics* **15**(2), 70–73 (1967). <https://doi.org/10.1109/TAU.1967.1161901>
10. Yurtman, A., Barshan, B.: Activity recognition invariant to sensor orientation with wearable motion sensors. *Sensors* **17**(8), 1838 (2017). <https://doi.org/10.3390/s17081838>